

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and

network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industryrelevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach

Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

Star Topology for a 20-Person Startup

A star topology is an ideal choice for a 20-person startup because all computers and network devices are connected to a central switch. This design provides good performance, easy management, reliability, and simple troubleshooting.

1. Central Switch

At the center of the network, a managed Layer 2/Layer 3 switch with at least 24 ports can be used. The 24 ports are sufficient for 20 employees' computers, printers, VoIP phones, and other devices, while also providing some extra ports for future expansion.

2. Network Cabling

Each workstation and network device is connected directly to the central switch using Cat6 UTP cables. Cat6 cables support Gigabit Ethernet and can be used for connections up to approximately 100 meters, which is sufficient for a typical office environment.

3. Router and Firewall

The central switch is connected to a router/firewall. The router provides access to the Internet and can perform important functions such as:

- Internet connectivity
- DHCP
- Network Address Translation (NAT)
- Basic network security
- Traffic management

4. Wi-Fi Access Point

A Wi-Fi access point can also be connected to the switch. It provides wireless network access to laptops, smartphones, tablets, and other mobile devices.

5. Easy Management

Star topology is easy to manage because all devices are connected to one central switch. The network administrator can monitor and configure connected devices from the central switch.

6. Fault Isolation

One of the biggest advantages is fault isolation. If the cable or network connection of one workstation fails, normally only that particular workstation is disconnected.

The other 19 employees can continue using the network without interruption.

7. Easy Troubleshooting

Troubleshooting is easier because each device has an individual connection to the switch. If a particular computer cannot access the network, the administrator can check its cable, switch port, or network configuration without affecting other users.

8. Scalability

The proposed 24-port switch provides room for future expansion. New computers, printers, IP phones, or other devices can be connected to available ports.

If more devices are required later, another switch can be added.

9. Centralized Cable Management

Since all connections terminate at the central switch, cable management becomes easier. This simplifies future moves, additions, and changes in the office network.

Mesh Topology for a Bank Network

A bank requires **near-zero downtime** because banking services and financial transactions must be available continuously. Therefore, a **full mesh or partial mesh topology** is suitable because it provides high reliability, fault tolerance, and multiple paths for communication.

1. Mesh Topology

In a mesh topology, network devices are connected through multiple links. A **full mesh** connects every important device directly to every other device.

For a bank, a **partial mesh** is more practical because only critical devices, such as core switches, routers, and firewalls, need multiple connections.

2. High Fault Tolerance

The main advantage of mesh topology is **fault tolerance**. If one network link fails, communication can continue through another available path.

Therefore, failure of a single link does not normally bring down the entire banking network.

3. Redundant Connections

The core network can have multiple connections between core switches and firewalls. Distribution and access switches can also have **redundant uplinks** to at least two core switches.

This prevents a single connection from becoming a point of failure.

4. Redundant Routers

Banks can use **two enterprise-grade routers** to provide redundant Internet connectivity.

If one router fails, the second router can continue providing network connectivity, helping to maintain continuous banking services.

5. Fiber Optic Cables

Fiber optic cables can be used between important core network devices.

Fiber provides:

- High bandwidth
- Low latency
- Resistance to electromagnetic interference
- Reliable long-distance communication

This is especially useful for financial transactions where fast and reliable communication is important.

6. Dynamic Routing

Routing protocols such as **OSPF or BGP** can be used to manage multiple paths.

When a link fails, the routing protocol can identify the failure and select an alternative path, allowing network communication to continue.

7. Redundant Power

Network reliability also depends on power availability. Therefore, the bank can use:

- Redundant power supplies
- UPS systems
- Backup generators

These systems help network devices continue operating during power failures.

8. Reliable Data Storage

Banks can use **RAID storage systems** to protect important data.

If one storage disk fails, the RAID system can continue providing access to the data depending on the RAID level used.

Bus Topology for a Small Classroom Lab

Bus topology is a suitable choice for a small classroom lab with a limited budget because all computers share a **single communication backbone cable**. It does not require an expensive central switch or server.

Main features:

1. **Low cost:** A single cable connects multiple computers, reducing hardware and installation costs.
2. **Simple setup:** Each computer can be connected to the backbone using connectors such as **T-connectors and BNC connectors** in a traditional coaxial setup.
3. **No dedicated server:** A teacher's computer can share files and other resources using built-in Windows/Linux file-sharing facilities.
4. **Suitable for light usage:** It works well for activities such as **internet browsing and basic file sharing** in a small classroom.
5. **Limitations:** The network is prone to **collisions**, and if the main cable breaks, communication for all connected computers can be affected.

Conclusion:

Therefore, bus topology is a **simple and economical solution** for a small classroom lab where the number of computers and network traffic are limited.

Hybrid Topology for a Large University Campus

A **hybrid topology** is suitable for a large university campus because it combines different network topologies to provide **high performance, reliability, and scalability**.

1. **Campus Backbone:** A **fiber-optic ring or mesh topology** is used at the backbone level to provide redundancy and prevent a single link failure from affecting the entire campus.
2. **High-Speed Connection:** High-speed **fiber-optic cables** connect the core routers and switches between different campus buildings.
3. **Building-Level Network:** Each building has a **distribution switch** connected to the campus core using dual fiber links. This provides reliable communication between buildings.

4. **Star Topology:** Inside each building, a **star topology** connects floor-level access switches to individual computers and devices. This makes the network easy to manage and expand.
5. **Scalability:** New buildings can be added by installing a new distribution switch and connecting it to the existing core network without disturbing the existing infrastructure.

Conclusion:

Thus, a hybrid topology provides **redundancy at the backbone, simplicity within buildings, high speed, easy management, and scalability**, making it ideal for a large university campus.

Network Design for a Company with Multiple Departments

A **hierarchical network architecture** using switches, routers, and VLANs is suitable for a company with multiple departments such as HR, Finance, Sales, and IT.

1. **Department-Level Switches:** Each department is connected to dedicated **Layer 2 switches**. These switches connect computers, printers, and other devices within the department.
2. **Layer 3 Core:** The department switches are connected to a central **Layer 3 switch or router**, which enables communication between different departments through **inter-VLAN routing**.
3. **VLANs:** Separate **VLANs** are created for each department. This improves security, reduces unnecessary broadcast traffic, and logically separates departmental networks.
4. **Internet and Security:** A router connects the internal network to the **Internet**. A firewall can be used to protect the company network from unauthorized access.
5. **Reliability and Scalability:** **Redundant links, backup paths, proper IP addressing, and subnetting** improve reliability and make the network easier to manage and expand.

Conclusion:

Thus, a hierarchical network with **Layer 2 switches, a Layer 3 core, VLANs, routers, and security measures** provides efficient communication, better security, high performance, and easy expansion for a growing company.